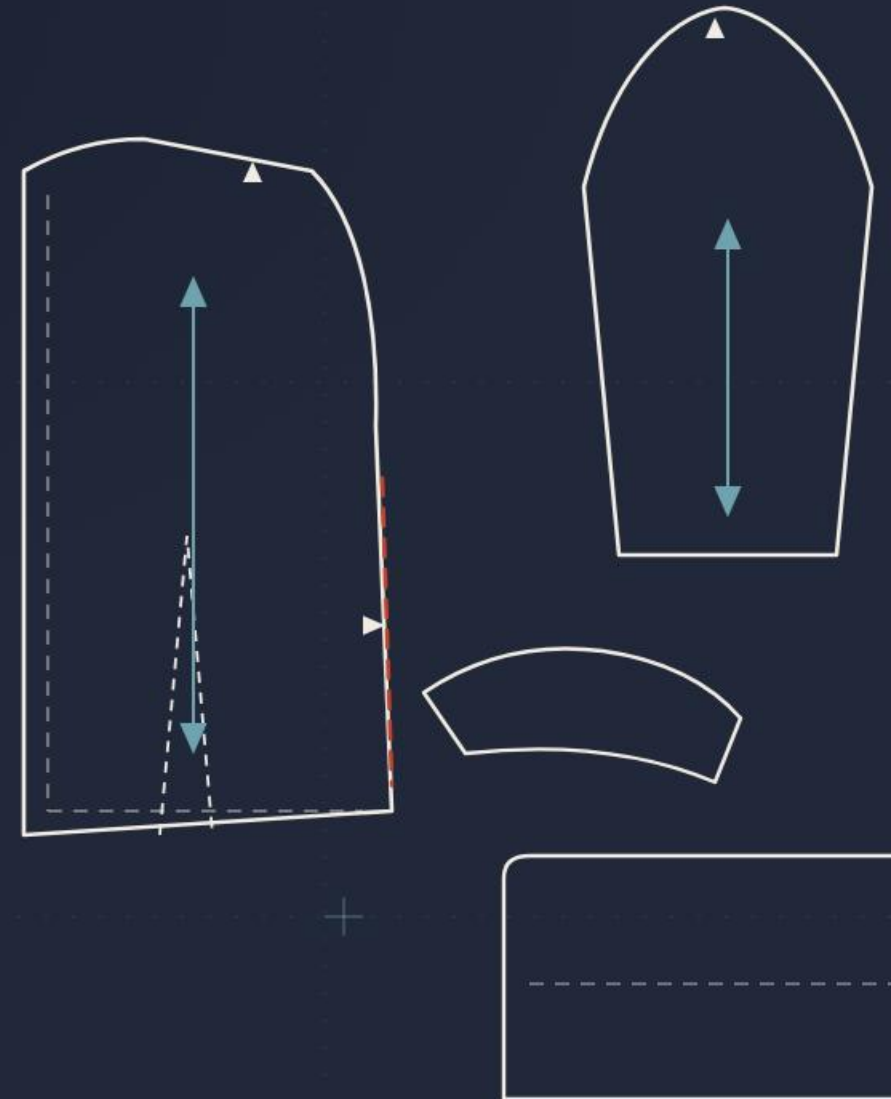
IEEE/CVF CVPR 2025

# AIpparel: A Multimodal Foundation Model for Digital Garments

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OUTLINE

# What we will cover

THE THROUGH-LINE **Problem → Method → Evidence** A method-then-evidence structure: we motivate the two enablers first, then let the numbers make the case.

| | | |
|---|---|---|
| 01 | **Motivation & prior limitations**<br>Why garment design is hard, and where single-modal methods fall short | PROBLEM |
| 02 | **AIpparel — method overview**<br>Fine-tuning a large multimodal model on garment sewing patterns | METHOD |
| 03 | **GCD-MM — the multimodal dataset**<br>A new large-scale garment corpus with paired text and images | METHOD |
| 04 | **Compact tokenization & regression**<br>A sewing-pattern token scheme with dedicated coordinate regression heads | METHOD |
| 05 | **Experiments & ablations**<br>Quantitative results across image, multimodal and editing tasks | EVIDENCE |
| 06 | **Takeaways & conclusion**<br>What it enables, current limitations, and the road ahead | EVIDENCE |

BACKGROUND

# Garment design is essential — and still largely manual

Existing sewing-pattern models are single-modal and struggle with complex geometry

## Clothing is fundamental to daily life and self-expression — yet a garment still begins with **slow, manual pattern-making**.

Designers translate intent into flat 2D panels that drape into a 3D shape; even expert tailors iterate on verbal and visual feedback.

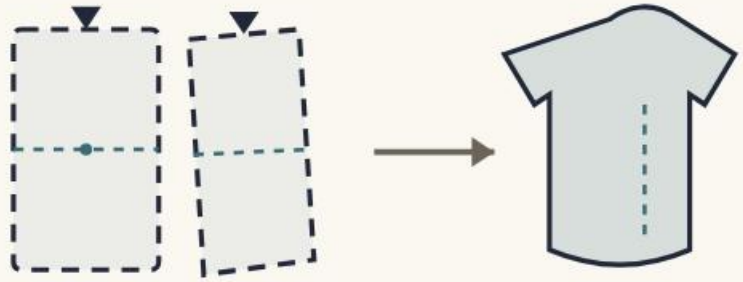
### PRIOR PATTERN-PREDICTION — SINGLE-MODAL

| | | |
|---|---|---|
| 3D points | → | task-specific model |
| Images | → | task-specific model |
| Language | → | task-specific model |

Hard to adapt to combined modalities

### THE GOAL — ONE SCALABLE MODEL

Language + Images → one multimodal model



Sewing pattern = flat 2D panels → 3D garment

### TWO BARRIERS TO A MULTIMODAL MODEL

1. No large-scale **multimodal** sewing-pattern dataset is publicly available
2. Accurate multimodal interpretation emerges only in **billion-parameter** models

THE GAP

# Prior methods are single-modal, data-starved, and geometry-limited

Figure 1 previews complex, simulation-ready patterns from text and images

01

### Single-modality focus

Prior models are task-specific — hard to adapt to combined text + image inputs.

02

### Data scarcity

No public large-scale multimodal sewing-pattern dataset with text, images and editing pairs.

03

### Complexity limits

The earlier multimodal attempt (Wang et al. [63]) was confined to simple garments with a fixed parameter set.




**Figure 1 in the paper (M1:p1)** Multimodal input → AIpparel → sewing pattern → direct 3D simulation

OVERVIEW

# One model, fine-tuned on a new dataset with a compact tokenizer

Three contributions: GCD-MM • a 100× more compact tokenization • a multimodal foundation model

THE RECIPE

Fine-tune a strong large multimodal model — **LLaVA 1.5-7B** — on a dataset purpose-built for garments.




01 DATASET

## GCD-MM

The first large-scale multimodal sewing-pattern dataset — pairing text, images and editing instructions.

**120,000⁺**

unique garments • ~10× larger than prior generative datasets

02 REPRESENTATION

## Compact tokenization

A novel tokenization scheme with a new training objective — a concise, learning-friendly pattern representation.

**100×**

fewer tokens than the DressCode tokenizer on the same garment

03 MODEL

## AIpparel

The first multimodal foundation model for sewing-pattern prediction — taking language, images and patterns as input.

**Enables**

language-instructed interactive editing of garment patterns

DATASET · CONTRIBUTION 1

# GCD-MM: the only large-scale dataset with text, images, and edits

Table 1 — modality comparison of sewing-pattern datasets

| DATASET | TOTAL | TEXT | IMAGE | EDITS |
|---|---|---|---|---|
| Wang et al. [63] | 8k | ✗ | ✓ | ✓ |
| Korosteleva & Lee [27] | 23.5k | ✗ | ✓ | ✗ |
| SewFactory [36] | 19.1k | ✗ | ✓ | ✗ |
| DressCode [21] | 20.3k | ✓ | ✗ | ✗ |
| GCD [30] | 130k | ✗ | ✓ | ✗ |
| **GCD-MM (OURS)** | **120k** | ✓ | ✓ | ✓ |

**Only complete row.** GCD-MM extends GarmentCodeData (GCD [30]) — image-only at 130k — by adding text captions and editing pairs, reaching all three modalities across ~120k garments.

Table 1 in the paper · p.3

DATASET CONSTRUCTION

# Building GCD-MM in three steps

Extend GCD → annotate text (rules + GPT-4o) → generate editing pairs

**01**

## Build on GCD

Extend **GarmentCode-Data** [30], the largest sewing-pattern corpus, with two new modalities.

- + text descriptions
- + editing pairs

**Forms GCD-MM**

**02**

## Annotate text

**Rule-based phrases**
type • features

+ rendered draped views

**GPT-4o [69]**

**Natural-language descriptions**

**Grounding on reliable phrases sharply cuts hallucination.**

**03**

## Generate editing pairs

Use the **GarmentCode** [29] programming abstraction to specify common edits.

- skirt / pants length
- necklines & inserts
- add / exclude hood or sleeve

**Each edit → paired pattern + templated instruction**

**Pipeline order.** Every stage feeds the next: the base corpus is annotated, then programmatically edited into instruction-paired samples.

Section 3.1

METHOD · CONTRIBUTION 2

# Sewing patterns as drawing-command tokens — ~100× more compact

~250 tokens on average (max 838) vs >30k for DressCode

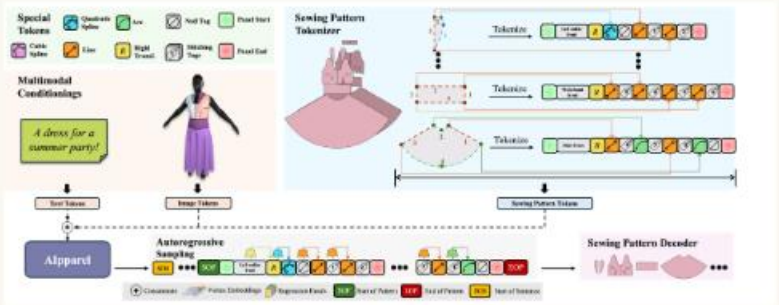



Figure 2 in the paper (M1:p4) — multimodal input → LLaVA → autoregressive tokens + regression heads → simulation-ready pattern

COMPACTNESS

**~100×**

fewer tokens than the DressCode tokenizer for the same garment

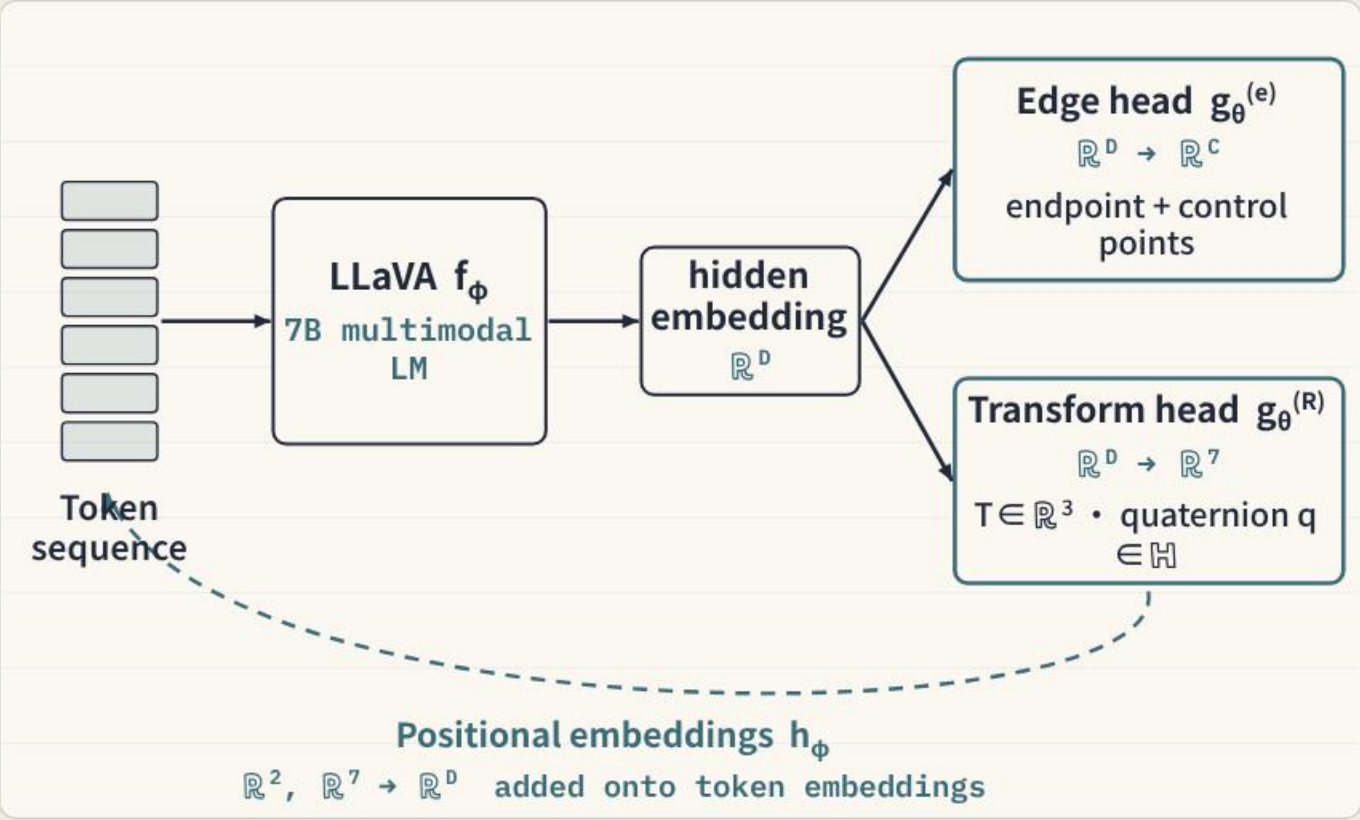
| | |
|---|---|
| ~250 | avg tokens per pattern (GCD-MM) |
| 838 | max tokens across the dataset |
| >30k | tokens per pattern for DressCode |

**Special-token vocabulary** A pattern becomes a short sequence of drawing commands wrapped in these tokens.

| | | | | |
|---|---|---|---|---|
| `<SoG>` `<EoG>` Garment start / end | `<SoP>` `<EoP>` Panel start / end | `<R>` Rigid transformation | `<L>` `<cL>` Line edge (open / closing) | `<Q>` `<cQ>` Quadratic spline |
| `<B>` `<cB>` Cubic Bézier | `<A>` `<cA>` Arc edge | `<t1>` … `<tM>` Stitching tags (paired edges) | `<tN>` Null tag (unstitched) | `LLaVA 1.5-7B` Base multimodal model |

# Regressing geometry instead of quantizing it

MLP regression heads + positional embeddings for vertices and 3D transforms




Section 3.2 · M1 p.5

Continuous geometry — vertex positions, control points, and 3D panel transforms — is **regressed, not quantized.**

- Prior work quantizes continuous parameters into discrete tokens, adding quantization error and sequence length.
- Small two-layer MLP heads read each edge and panel transform straight from the preceding hidden embedding.
- Positional-embedding MLPs project those values back into $\mathbb{R}^D$ and add them onto the token embeddings.

**Train** with ground-truth values · **generate** with regressed values

TRAINING DETAILS

# One model on 120k garments, fine-tuning LLaVA 1.5-7B

Frozen vision encoder • CE + L2 objective • 90:5:5 split

**120,000+**
UNIQUE GARMENTS • GCD-MM
Multimodal annotations • one training corpus

**LLaVA 1.5-7B**
BASE MULTIMODAL MODEL
Fine-tuned end-to-end for garment generation

FROZEN

- ▢ Vision encoder
- ▢ Vision–language projection

TUNE

FINE-TUNED

- Language-model weights
- Regression heads — edge & panel
- Positional-embedding MLPs

OBJECTIVE Eq. 2 • § 3.2, 4

**Cross-Entropy** discrete **+ weighted L2** continuous

edge endpoints & panel transforms

DATA SPLIT

**90 / 5 / 5**

train / val / test

COVERAGE

**One model, all tasks**

image • text • editing — all GCD-MM results

EXPERIMENTAL SETUP

# Three tasks, two baselines, geometry + stitch metrics

Image-to- / text-to-garment and editing • vs SewFormer & DressCode

## 01 TASKS

- **Image → garment**
- **Text → garment**
  multimodal generation
- **Language-instructed editing**
  edit an existing garment

## 02 BASELINES

**SewFormer** [36]

**DressCode** [21]

Single-modal — augmented with off-the-shelf LMMs:

| | |
|---|---|
| SewFormer-FT † / DressCode † | MULTIMODAL |
| SewFormer* / DressCode* | EDITING |

via GPT-4o • DALL • E • InstructPix2Pix

## 03 METRICS

↓ **Geometry L2** lower better • cm*

Panel L2 | Rot L2 | Transl L2

↑ **Accuracy** higher better • %

#Panel | #Edge | #Stitch

*L2 in centimeters, except rotation

COMPACT TABLES **Accuracy = #Panel Acc × #Edge Acc** Metric names, units & directions reused on Tables 2–5

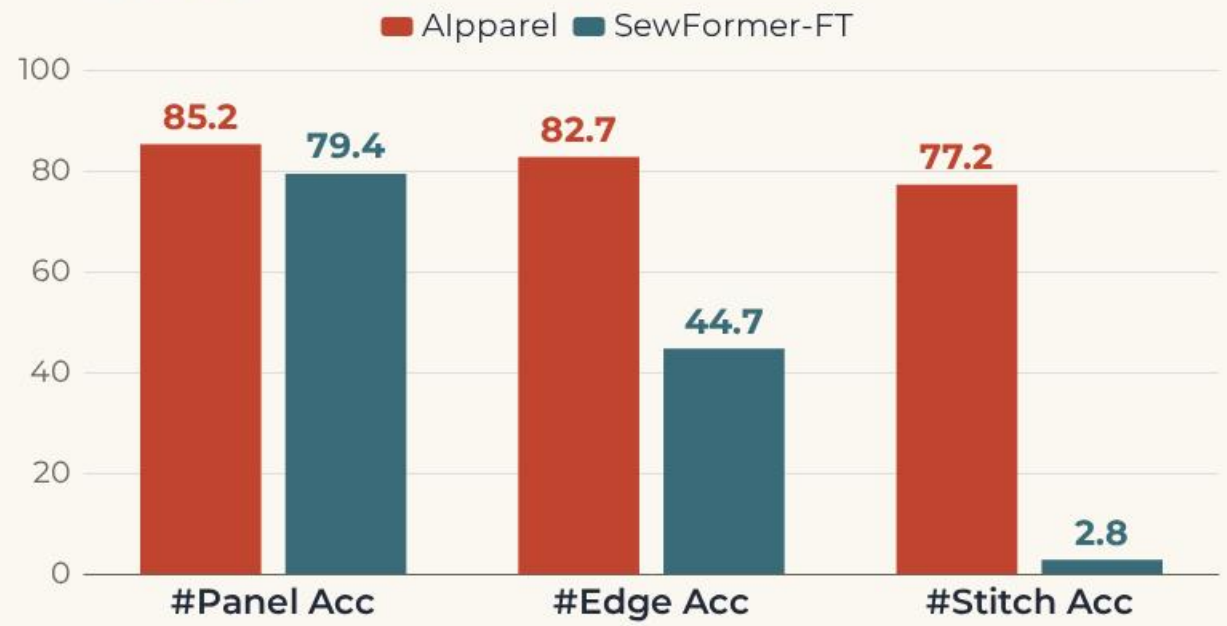
Section 4 (Metrics), M1:p6

# State of the art on both datasets — a large margin on GCD-MM

#Stitch Accuracy 77.2% vs 2.8% on the hard GCD-MM data

## GCD-MM accuracy — AIpparel vs fine-tuned SewFormer

Accuracy (%), higher is better




Source: Table 2 — Image-to-Garment Prediction

### Also on GCD-MM • Panel L2 error (cm, lower is better)

| | |
|---|---|
| AIpparel | **5.4** |
| SewFormer-FT | **12.3** |

Translation L2 also halved: 2.7 vs 4.5 cm

### Easier SewFactory benchmark — AIpparel leads across the board

**2.8 cm**
Panel L2 error (vs 3.3)

**99.8 %**
#Stitch Acc (vs 99.2)

# Directly handling multimodal, common-sense queries

Table 3 — Accuracy 59.0 vs 10.3 (SewFormer-FT †) and 0.6 (DressCode †)

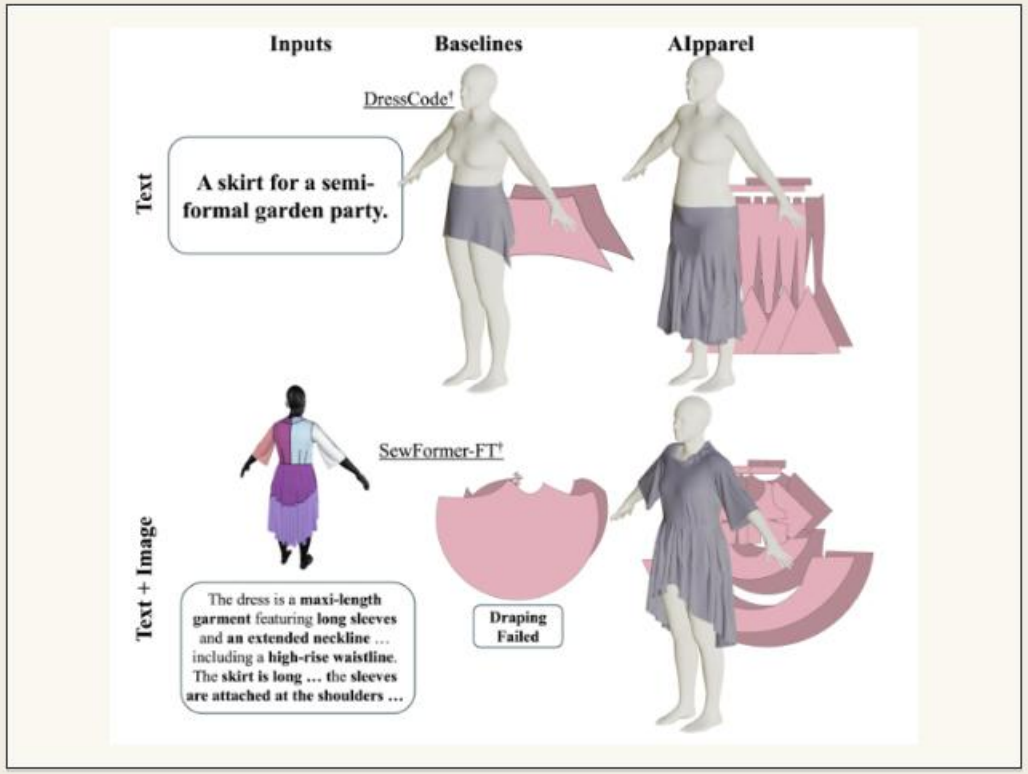



Figure 4 in the paper — common-sense text query (top) and combined image + text (bottom)

Baselines are augmented with GPT-4o / DALL · E, yet still fail; AIpparel answers the query directly.

| METHOD | ACCURACY ↑ | PANEL L2 ↓ |
|---|---|---|
| SewFormer-FT † | 10.3 | 22.4 |
| DressCode † | 0.6 | 31.0 |
| **AIpparel** | **59.0** | 6.1 |

SETUP 100 garments · 5 input types × 20 samples — text, image, text + image, natural-language queries, editing.

Figure 4 & Table 3 in the paper · M1:p7

RESULTS · EDITING

# Precise, style-preserving edits from language

Table 4 — Accuracy 83.4 and Panel L2 1.5 cm, far ahead of baselines

**TASK** Edit an existing pattern from a text instruction — without changing overall style. Baselines adapted for editing via InstructPix2Pix and GPT-4o.

| METHOD | ACCURACY ▲ | PANEL L2 ▼ CM |
|---|---|---|
| SewFormer* | 9.5 | 18.6 |
| DressCode* | 37.0 | 13.7 |
| **AIpparel** | **83.4** | **1.5** |

* baselines adapted for editing; not native editors.

Table 4 in the paper

AIPPAREL VS. ADAPTED DRESSCODE

**+46.4**
Accuracy points

**−12.2 cm**
Panel L2 error

AIpparel edits the pattern directly, making only the requested change. Baselines translate the instruction into keywords and lose the garment's original style.

VISUAL ANALYSIS

# Fine details preserved; edits stay on-style

Figure 3 — reconstruction quality • Figure 5 — editing precision




**Fig. 3 • Reconstruction**

On hard GCD-MM inputs SewFormer fails to drape at all, while AIpparel recovers even the small panels — waistbands and sleeve cuffs.

**The qualitative evidence behind the metrics: AIpparel stays faithful to both garment geometry and the editing instruction.**

Figure 3 (M1:p5) • Figure 5 (M1:p7)

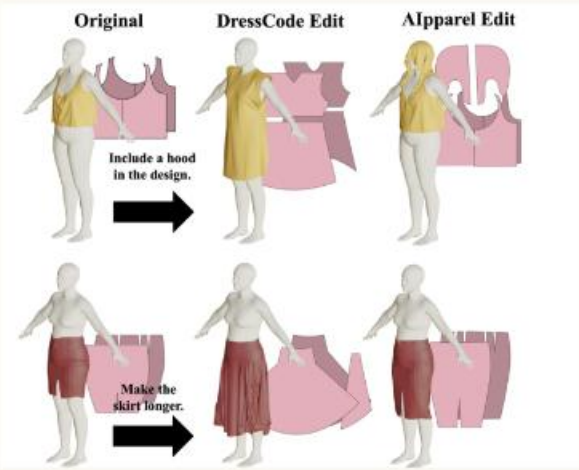



**Fig. 5 • Editing**

Add a hood or lengthen a skirt: AIpparel changes exactly that, while DressCode over-edits and drifts off-style.

ABLATION

# Both the tokenizer and the mixed objective matter

Table 5 — Accuracy 38.4 → 79.0 → 85.0, with a >25× inference speed-up

| TEXT-TO-GARMENT • DRESSCODE BACKBONE | Accuracy ↑ | Panel L2 cm ↓ | Time s ↓ |
|---|---|---|---|
| **DressCode** baseline tokenizer + CE | 38.4 | 22.4 | 52.2 |
| **Ours** w.o. reg. our tokenizer + next-token | 79.0 | 7.2 | 3.4 |
| **Ours** our tokenizer + mixed objective | 85.0 | 6.1 | 2.1 |

Table 5 (in the paper) • averaged over 100 test samples • wall-clock on a single A6000 GPU

**+ New tokenizer**

**38.4 → 79.0**

Reconstruction accuracy more than doubles.

**+ Mixed regression objective**

**79.0 → 85.0**

Further gain; panel error drops 7.2 → 6.1 cm.

**>25× faster**

52.2s → 2.1s per garment

# Compact tokens scale LMMs to garments — with real bounds

Multimodal training enables intuitive editing; manifold-only geometry and synthetic data remain limits

What the work teaches

## Takeaways

01 **Compact tokenization is the enabler.**
Efficient tokenization is what lets large multimodal models scale to complex garment geometry.

02 **Multimodal training unlocks interaction.**
Genuinely multimodal data enables intuitive design and interactive, conversational editing.

Where it stops — stated candidly

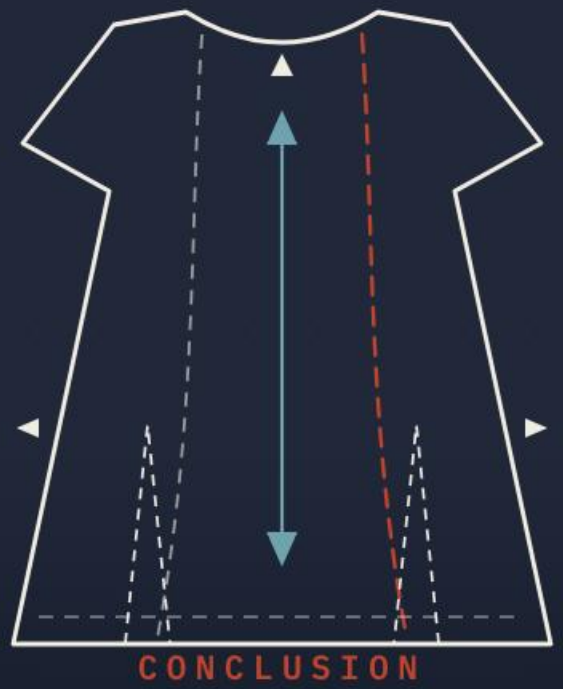
## Limitations

01 **Manifold-only geometry.**
Covers garments as manifold surfaces; features like pockets need non-manifold structure.

02 **Synthetic data.**
Training relies on procedurally generated garments, not real-world capture.

03 **Fabrication.**
Producing real, material-feasible garments under physical constraints is future work.

**Next:** extend to non-manifold features and to fabrication under real physical constraints. Section 5




# The first multimodal foundation model for digital garments

A scalable recipe — SOTA single- and multi-modal generation, plus language-driven editing. GCD-MM to be released.

georgenakayama.github.io/AIpparel